

Hafiz Rehan Ahmad

MERN Stack Developer

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Education

Faislabad, Pakistan	Govt College University	Oct 2017 – Oct 2021
- I pursued Bachelor of Science in Computer Science (BSc CS) from GC University. - Acquiring in-depth knowledge in software development, algorithms, and data structures Gaining hands-on experience through coursework and ongoing projects. - Developing proficiency in programming languages and software tools Participating in team-based assignments to enhance collaboration and communication skills. - Engaging in research and analysis of current software engineering trends and technologies		
Gujranwala, Pakistan	Be Superior College	Oct 2016 – Aug 2017
- Pursued Intermediate of Computer Science (ICS) from Be Superior college. - Developed proficiency in C programming language through coursework and projects. - Completed multiple programming assignments and practical labs, showcasing problem-solving skills. - Engaged in collaborative projects to enhance teamwork and project management abilities. Demonstrated strong analytical skills through rigorous coursework and exams.		

Experience

Wapda Town Gujranwala, Pakistan	Appcrates	May 2024 – Present
Mern Stack Developer		
- Designed and developed a real-time map tracking system using the MERN stack to monitor and manage fleet operations effectively. Frontend Development: - Built a responsive user interface using React.js, incorporating features like live tracking, device management, and route visualization. - Utilized React hooks and Redux Toolkit for efficient state management and seamless integration of map overlays. - Integrated dynamic map layers and markers using Leaflet.js and Mapbox APIs, enabling real-time updates of vehicle locations. - Implemented advanced filtering and search functionality for devices, geofences, and historical trip data. Backend Development: - Developed RESTful APIs with Node.js and Express.js to handle data fetching, route optimization, and user authentication. - Designed a robust database schema in MongoDB to manage vehicle positions, trip histories, and geofence configurations. - Integrated WebSocket communication to provide real-time updates of vehicle positions and alert notifications. Key Features Delivered: - Real-time vehicle tracking with route playback functionality and speed overlays.		

- Geofencing capabilities, allowing users to create and manage custom zones with entry/exit alerts.
- Historical trip analysis with detailed metrics like distance traveled, stops, and average speed.
- User roles and permissions, enabling access control for admins, managers, and end-users.
- **Performance Optimization:**
- Implemented database indexing and query optimization, reducing response time by 40% for high-traffic queries.
- Improved frontend performance by introducing **lazy loading** for components and map tiles.

Lahore, Pakistan
Mern Stack Developer

Hexa Tech Solution

March 2023 – April-2024

- Built scalable web applications using the MERN stack with optimized MongoDB models and secure RESTful APIs.
- Developed responsive UIs with React, Redux, and modern CSS frameworks like Material-UI and Tailwind CSS.
- Handled server-side logic, integrated third-party services, and implemented JWT/OAuth2 authentication.
- Deployed applications to AWS/Heroku with CI/CD pipelines and managed server environments.
- Conducted comprehensive testing using Jest and Cypress, ensuring high reliability and performance.
- Collaborated on projects with Git, Agile practices, and code reviews for quality assurance.

Tecno Sphere, Gujranwala
React JS Developer

Tecno Sphere

Jan2022 – Feb-2023

- Developed modern web applications exclusively using **React functional components** and **React hooks**.
- Implemented state management with **Redux Toolkit** and optimized component rendering using **useMemo** and **React.memo**.
- Created reusable, modular components to ensure maintainable and scalable codebases for complex UI designs.
- Leveraged **useEffect** for side effects and asynchronous data fetching, ensuring efficient component lifecycle management.
- Built responsive and accessible UIs using **Material-UI** and **Tailwind CSS**, adhering to WCAG accessibility standards.
- Integrated **RESTful APIs** and enhanced user interactivity by implementing features like infinite scroll, search, and filtering.
- Improved performance by introducing **lazy loading** and **code-splitting** via **React.lazy** and **Suspense**.
- Refactored legacy class components into functional components, reducing code complexity by 30%.
- Developed custom hooks for reusable logic, such as API calls, form handling, and input validation.
- Built dynamic, interactive SPAs using **React functional components** with a focus on reusable design patterns.

- Employed **useState** and **useReducer** hooks for efficient state management in complex forms and multi-step workflows.
- Integrated with third-party services like Google Maps and Chart.js, enabling dynamic and data-driven UI updates.
- Ensured performance optimization by reducing unnecessary renders using **React.memo** and selective re-rendering strategies.
- Delivered pixel-perfect UIs by collaborating with designers and implementing Figma designs using functional components.
- Worked with **Formik** and **Yup** for advanced form handling and validation, streamlining the user input process.

Skills

Category	Skills
Programming Languages	JavaScript, Typescript
Front End	HTML, CSS, JavaScript, ReactJS, Next jsTailwinds CSS
Backend	Nodejs, ExpressJS
Databases	MongoDB, MySQL
Dev Tool	VsCode, Postman,

Projects

MarketPlace

Backend + Frontend (NodeJS, ReactJS)

- Designed and developed a scalable **eCommerce marketplace platform** using the **MERN stack**, enabling multi-vendor functionality for seamless product listing and transactions.
- **Frontend Development:**
 - Built a dynamic and responsive UI with **React.js**, implementing features like product catalogs, advanced search, and filtering.
 - Integrated real-time updates for inventory, cart, and order tracking using **React hooks** and **Redux Toolkit**.
 - Developed custom components for user dashboards, product reviews, and order management.
 - Enhanced user experience by integrating third-party libraries like **Material-UI** and **Tailwind CSS** for styling.
- **Backend Development:**
 - Developed RESTful APIs with **Node.js** and **Express.js** to manage product listings, user authentication, and order processing.
 - Implemented **JWT-based authentication** and role-based access control for vendors, customers, and administrators.
 - Designed and optimized **MongoDB** schemas for products, users, orders, and payments, ensuring data integrity.

Tabae Tracking System, Saudi Arabia

- Developed the front-end interface for a **real-time tracking system** to monitor vehicles, assets, and personnel across Saudi Arabia.
- **UI Development:**
- Built dynamic and responsive user interfaces using **React.js**, ensuring seamless user experiences across devices.
- Designed interactive maps with **Leaflet.js** and **Mapbox APIs**, enabling real-time tracking and geofence management.
- Implemented multi-language support (Arabic and English) with **react-i18next** for localization.
- Created reusable components for dashboards, trip history playback, and geofence visualization.
- **State Management:**
- Used **React hooks** and **Redux Toolkit** to manage state efficiently across the application.
- Designed robust state workflows for handling live map updates, user interactions, and alert notifications.
- **Performance Optimization:**
- Optimized rendering performance with **React.memo**, **lazy loading**, and **code-splitting** for large datasets.
- Reduced load times by implementing pagination and infinite scroll for historical data and logs.
- **Key Features Delivered:**
- Real-time vehicle tracking with route visualization and status updates.
- Geofence management tools, including the ability to create, edit, and visualize geofences on the map.
- User-friendly dashboards displaying trip analytics, alerts, and activity summaries.
- Search and filter functionalities for devices, routes, and alerts.
- **Collaboration & Tools:**
- Worked closely with backend developers to integrate REST APIs for vehicle data, trip history, and notifications.
- Styled the application using **Tailwind CSS** for consistent and responsive designs.

Cricap Web

Frontend (NextJS)

- Developed the front end of the CRICAP website using the latest React.js framework.
- Implemented Server-Side Rendering (SSR) for optimized data fetching and improved SEO
- Integrated social media sharing functionality for enhanced user engagement
- Utilized Web Sockets to provide real-time updates on match odds.
- Deployed the application on AWS, ensuring scalability and reliability.
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Driving Test Management System Full Stack

Mern Stack

- Led the development of a **Driving Test Management System** using **MERN stack** with **MySQL** as the database, integrating **Sequelize** ORM for seamless database interaction and efficient data management.

- **Frontend Development:**

- Designed and implemented a responsive user interface using **React.js**, allowing candidates, instructors, and administrators to easily interact with the system.
- Created dynamic components for real-time test scheduling, result dashboards, candidate registration, and performance feedback.
- Utilized **Redux Toolkit** for state management across the application to ensure smooth user experience and reduce unnecessary renders.
- Styled the application with **Tailwind CSS** for a clean, responsive design.

- **Backend Development:**

- Developed RESTful APIs using **Node.js** and **Express.js**, handling candidate data, scheduling, and evaluation submissions.
- Integrated **JWT-based authentication** for secure access control, ensuring that users had appropriate permissions based on their roles.
- Designed and implemented endpoints for test scheduling, availability updates, result processing, and notifications.

- **Database Management:**

- Modeled and optimized **MySQL** relational database schemas using **Sequelize ORM** to manage data across multiple tables (e.g., candidates, instructors, test schedules, and results).
- Created complex queries and **Sequelize models** to fetch and update test-related data efficiently.
- Ensured data consistency by enforcing relationships (e.g., foreign key constraints) and optimized query performance using **indexes** and **Joins**.
- Built **stored procedures** and **transactions** to manage bulk data operations, such as batch updates or test results.